

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2024

Bachelor in Computer Applications Full Marks: 60
Course Title: Operations Research Pass Marks: 24
Code No: CAOR451 Time: 3 hours
Semester: 8

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions. [10×1 = 10]

1. Circle (O) the correct answer.

i) What is the primary goal of Operations Research?

- a) Increase confusion b) Optimize decision making
- c) Reduce profits d) Complicate problems

ii) Which method is used to find the optimal solution in linear programming?

- a) Binary search b) Simplex method
- c) Bubble sort d) Quick sort

iii) In LPP, the objective function is:

- a) Always minimized b) Maximized or minimized
- c) Always constant d) Never defined

iv) What is a feasible region in linear programming?

- a) Area outside constraints b) Area satisfying all constraints
- c) Only the origin d) A single point

v) Which of the following is a network model in OR?

- a) Linear programming b) Transportation problem
- c) Queuing theory d) Inventory model

vi) What does CPM stand for in project management?

- a) Critical Path Method b) Complete Project Management
- c) Computer Process Model d) Cost Performance Metric

vii) In queuing theory, what does λ represent?

- a) Service rate b) Arrival rate

- c) Queue length d) Server count
- viii) Which method solves assignment problems?
- a) Simplex method b) Hungarian method
- c) Northwest corner d) MODI method
- ix) What is a dummy activity in PERT/CPM?
- a) Real work b) Imaginary activity showing dependency only
- c) Delayed task d) Parallel task
- x) In game theory, a saddle point is:
- a) Maximum of row minimums = Minimum of column maximums b) Highest payoff
- c) Lowest payoff d) Random point

Group B

Attempt any SIX questions. [6×5=30]

2. Define Operations Research. Explain the scope and applications of OR in business and industry.
3. What is linear programming? Formulate a general linear programming problem.
4. Solve the following LPP graphically: Maximize $Z = 3x + 2y$ subject to $x + y \leq 4$, $x \leq 3$, $y \leq 3$, $x, y \geq 0$.
5. Explain the Simplex method for solving linear programming problems.
6. What is the transportation problem? Explain the Northwest Corner method for finding initial basic feasible solution.
7. Describe the Critical Path Method (CPM). How is it different from PERT?
8. What is queuing theory? Explain the basic components of a queuing system.

Group C

Attempt any TWO questions. [2×10=20]

9. Explain the assignment problem and the Hungarian method for solving it. Solve the following assignment problem to minimize cost:
 Person A: [10, 12, 15, 18]
 Person B: [14, 11, 13, 16]
 Person C: [12, 14, 10, 15]
 Person D: [16, 15, 12, 11]
10. What is game theory? Explain two-person zero-sum games. Solve the following game using maximin-minimax principle and check for saddle point:
 Player A strategies: [3, -1, 2; -2, 4, 1; 1, 0, 3]

11. Explain inventory control models. Derive the Economic Order Quantity (EOQ) formula. What are the assumptions of the EOQ model? How does it help in inventory management?